Day 086 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Solve Constant-Speed Problems

Standards: 6.RP.A.3b

Learning target: I can solve constant-speed problems using a rate per hour.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating.

Worked example 1

A bus travels at 40 km/h. How long does 280 km take?

1. At a constant speed, distance = speed × time.

2. Divide distance by time to find speed, or distance by speed to find time.

3. Time = distance ÷ speed = 280 ÷ 40 = 7.

4. State the result in the form requested: 7 hours.

Answer: 7 hours

Worked example 2

A bicycle travels 180 km in 6 hours at constant speed. Select all true statements.
A. The speed is 186 km per hour.
B. In 5 hours it travels 150 km.
C. The speed is 30 km per hour.
D. In 1 hour it travels 31 km.

1. At a constant speed, distance = speed × time.

2. Divide distance by time: 180 ÷ 6 = 30 km/h.

3. Multiply this unit rate by 5 for the later distance.

Answer: B, C - In 5 hours it travels 150 km.; The speed is 30 km per hour.

Day 086 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A cyclist travels 60 km in 6 hours at constant speed. Find the speed.

02 / Constructed response

A bus travels at 20 km/h. How long does 140 km take?

03 / Constructed response

A cyclist travels 210 km in 7 hours at constant speed. Find the speed.

04 / Constructed response

A bus travels at 10 km/h. How long does 50 km take?

05 / Constructed response

A cyclist travels 200 km in 5 hours at constant speed. Find the speed.

06 / Constructed response

A bus travels at 40 km/h. How long does 200 km take?

07 / Constructed response

A cyclist travels 45 km in 3/2 hours at constant speed. Find the speed and the distance in 5 hours.

08 / Multiple select

A bicycle travels 50 km in 2 hours at constant speed. Select all true statements.
A. The speed is 52 km per hour.
B. In 5 hours it travels 125 km.
C. The speed is 25 km per hour.
D. In 1 hour it travels 26 km.

09 / Multiple select

A bicycle travels 40 km in 4 hours at constant speed. Select all true statements.
A. The speed is 44 km per hour.
B. In 5 hours it travels 50 km.
C. In 1 hour it travels 11 km.
D. The speed is 10 km per hour.

10 / Constructed response

A bus travels at 25 km/h. How long does 25 km take?

Day 086 | Unit 06 | Answer key

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1. 10 km/h

Speed = 60 ÷ 6 = 10; distance = speed × time.

6.RP.A.3b | Constructed response

2. 7 hours

Time = distance ÷ speed = 140 ÷ 20 = 7.

6.RP.A.3b | Constructed response

3. 30 km/h

Speed = 210 ÷ 7 = 30; distance = speed × time.

6.RP.A.3b | Constructed response

4. 5 hours

Time = distance ÷ speed = 50 ÷ 10 = 5.

6.RP.A.3b | Constructed response

5. 40 km/h

Speed = 200 ÷ 5 = 40; distance = speed × time.

6.RP.A.3b | Constructed response

6. 5 hours

Time = distance ÷ speed = 200 ÷ 40 = 5.

6.RP.A.3b | Constructed response

7. 30 km/h; 150 km

Speed = 45 ÷ 3/2 = 30; distance = speed × time.

6.RP.A.3b | Constructed response

8. B, C - In 5 hours it travels 125 km.; The speed is 25 km per hour.

Divide distance by time: 50 ÷ 2 = 25 km/h. Multiply this unit rate by 5 for the later distance.

6.RP.A.3b | Multiple select

9. B, D - In 5 hours it travels 50 km.; The speed is 10 km per hour.

Divide distance by time: 40 ÷ 4 = 10 km/h. Multiply this unit rate by 5 for the later distance.

6.RP.A.3b | Multiple select

10. 1 hours

Time = distance ÷ speed = 25 ÷ 25 = 1.

6.RP.A.3b | Constructed response